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H.W

- Q) Find out the Applications of Recursion
- Q) Is recursion still used in the present programming era. If so, where?
- Q) What are the prog. lang. that support recursive calls.

1. A) Recursion is ideal for problems that can be broken into smaller, self-similar sub-problems. Applications include:

→ Sorting & Searching: Algorithms like Quick Sort, Merge Sort, and Binary Search.

→ Backtracking: Solving puzzles (Sudoku, N-Queens) and pathfinding in games.

→ Tree / Graph Traversal: Exploring nested structures (like file directories or DOM trees) via DFS (Depth-first Search).

→ Mathematical Functions: Calculating Factorials, Fibonacci series, and GCD.

2. A) Yes, Recursion is essential. It is the core of functional programming paradigms, used heavily in compiler Design (parsing

nested code structures), and utilized ²³ modern data processing libraries to handle Hierarchical data.

3. A) Recursion is a standard feature in nearly Almost all Modern Languages

C, C++, Java, Python, JavaScript, Go & Rust.